

Problem Set 1

ECE 455 / ECE 555, Fall 2026

Assigned	Wednesday, September 2
Due	Friday, September 11, 11:59 p.m. , through Canvas

How to submit

One PDF. Handwritten and scanned is fine if it is legible. Code may be typed or handwritten. Show your arithmetic; an answer with no work shown earns partial credit at best.

AI Use Statement

Generative AI tools are permitted, but their use must be disclosed. End your report with a short AI Use Statement naming the tool, what you used it for, and what it contributed. If you used no AI tool, write "No generative AI tools were used."

Late Work

Submissions are due at 11:59 p.m. on the stated date. **Late submissions are not accepted.** If you must miss a submission date, inform the instructor in advance, or submit a legitimate, documented reason for the absence. Failure to do so results in a grade of zero for this assignment. See the syllabus for the full policy.

1. Bit-specific definitions (11 points)

- a) Fill in the table with the bit-specific C definitions. Use the APB aperture, whose Port A base address is 0x40004000. The first row is done for you.

Port	Bits	Definition
A	5	<code>#define PA5 (*((volatile uint32_t *)0x40004080))</code>
A	1	
A	3, 1, 0	
A	6, 3, 2, 0	
A	6, 4, 1	
D	7, 6, 5	
F	7, 0	

- b) For bit 6, 3, 2, 0 of PortA, show how you computed the address from the port base and the bit mask.

2. Friendly initialization and two square waves (13 points)

- a) Write C that initializes TM4C Port A in a friendly manner so that pins 0, 2 and 3 are outputs and the rest are unchanged inputs. Do not use bit-specific addressing.
- b) Using that port, generate two square waves on PA2 and PA1 that are 180 degrees out of phase. Do not use bit-specific addressing.
- c) State what "friendly" means in one sentence and point to the line in your code that makes it friendly.

3. A repeating LED pattern, and its timing (31 points)

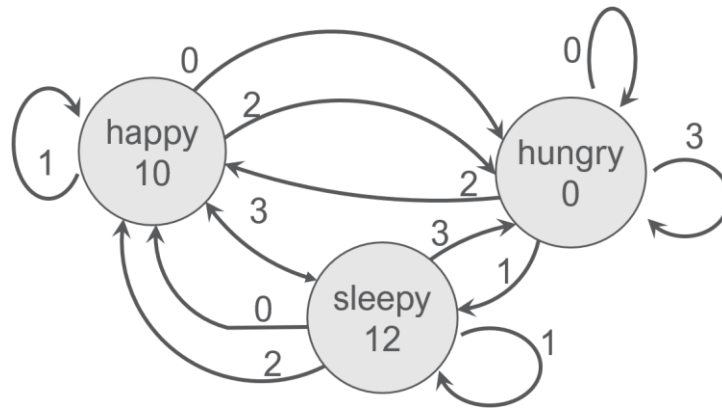
- a) Design an embedded system that flashes LEDs in the repeating binary patterns 1101, 1110, 1010, 1001, separated by 3 ms. Use only Port E. The bus clock is 80 MHz.

Requirements:

- Port E must be initialized.
 - Use bit-specific addressing for the output.
 - Use SysTick_Wait to insert the delay. State the value you pass to it and show the arithmetic that produced it.
- b) Using your own code from a), draw the timing diagram of PE3, PE2, PE1 and PE0 for two full cycles of the pattern. Label the time axis with real units.
- c) State the period of the full four-pattern cycle and the duty cycle of PE0.
- d) State the resolution with which you know each edge time, and say where that resolution comes from.

4. A Moore finite state machine (33 points)

Implement the Moore machine shown below in C. It has three states named happy, hungry and sleepy, and the initial state is happy. The input is on Port A bits 1 and 0. The output is on Port B bits 4, 3, 2, 1 and 0.



- Show the ROM-based data structure as an array of structs. Give the struct definition and the fully initialized array.
- Show the initialization and the controller. Initialize the direction registers, and make all port code friendly. Local variables may live wherever is appropriate. The repeating execution sequence is output, input, next.
- Take three arrows from the state graph given above. For each, name the exact element of your struct array in a) that encodes it.

5. Vocabulary (7 points)

In 16 words or less each, define:

- a) Latency
- b) Real time
- c) Bandwidth
- d) I/O bound
- e) CPU bound
- f) Blind-cycle synchronization
- g) Busy-wait synchronization

6. The periodic interrupt (5 points)

A student wants LED PD0 to toggle every 5 ms using the SysTick periodic interrupt, at a bus clock of 80 MHz. They write the following, adapted from the book's own pattern (Port D setup and the priority line are omitted here; assume PD0 is already configured as a friendly output):

```
void SysTick_Init(uint32_t period){
    NVIC_ST_CTRL_R = 0;           // disable SysTick during setup
    NVIC_ST_RELOAD_R = period - 1; // reload value
    NVIC_ST_CURRENT_R = 0;        // any write to current clears it
    NVIC_ST_CTRL_R = 0x00000007;  // enable with core clock and interrupts
    EnableInterrupts();
}

void SysTick_Handler(void){
    PD0 ^= 0x01;    // toggle PD0
}
```

- a) Compute the value of period that produces a 5 ms toggle. Show the arithmetic. (3 points)
- b) 0x00000007 sets three bits of NVIC_ST_CTRL_R at once. Name the three bits and state what each one does. (1 point)
- c) The file startup.s in your own project already has an entry in its vector table for SysTick_Handler, before this code is written. In one or two sentences, say why no vector-table edit is needed for this ISR to run. (1 point)

